Research-Backed Brief: The Future of EdTech in Ghana

The Future of AI in Education and Career Development in Ghana

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Introduction

Artificial intelligence (AI) is becoming an important part of education and career development. AI tools can help students understand difficult topics, improve their skills, and explore different career opportunities. In Ghana, these tools could help students and young people prepare for the job market. CareerGhana could also explore ways to use AI to improve its career guidance and learning services. However, it is important to consider challenges such as internet access, data privacy, and the accuracy of information provided by AI tools.

What the Research Shows

A study by Segbenya et al. (2023) looked at how postgraduate students in Ghana use AI and how it relates to their employability skills. The researchers collected data from 294 postgraduate students. They found that tools such as ChatGPT and QuillBot were commonly used for learning and academic work. The study also found that AI skills and were significant predictors of employability. This suggests that AI tools may help students develop skills that are useful in the workplace, although not guaranteeing employment.
Source: *Artificial Intelligence in Higher Education: Modelling the Antecedents of Artificial Intelligence Usage and Effects on 21st Century Employability Skills Among Postgraduate Students in Ghana* by Moses Segbenya, Brandford Bervell, Evans Frimpong-Manso, Isaac Christopher Otoo, Tahir Ahmed Andzie and Sampson Achina, 2023.

Another study by Asamoah and Amarteifio (2025) examined the challenges of using AI tutoring systems in Ghanaian higher education. These systems can help students learn by providing support based on their individual needs. However, there are challenges that must be addressed before they can be used effectively. This shows that although AI tutoring has potential, educational institutions need to consider the problems that may affect its use.
Source: *Overcoming Barriers and Ethical Challenges in Implementing Intelligent Tutoring Systems in Ghanaian Higher Education* by Moses Kumi Asamoah and Jessica Amarteifio, 2025.

Again, in a 2023 publication, UNESCO discussed the benefits and risks of using generative AI such as Google Gemini, ChatGPT, Microsoft Copilot etc. in education and research. The publication explained that AI can support learning and research, but it also raised concerns about privacy, fairness, and the

reliability of AI-generated information. It recommended responsible use of AI such that humans remain involved in important decisions. This means that AI should support students and educators rather than completely replacing human guidance.
Source: *Guidance for Generative AI in Education and Research* by Fengchun Miao and Wayne Holmes, 2023.

What This Means for CareerGhana

The research suggests several opportunities CareerGhana could explore by harnessing different ways of using AI to support students and job seekers. For instance, CareerGhana could consider using AI to help in career guidance; a platform where users explore career options based on their interests, skills, and educational backgrounds. This could make it easier for young people to understand the career paths available to them.

Also, the company could explore AI-assisted learning resources and tools that help users identify skills they need for their chosen careers and find relevant training opportunities. This could help students prepare for the job market.

The company should also consider responsible AI use. Information provided by AI tools should be checked for accuracy, and users' personal information should be protected and learners should be encouraged to verify important information. The company should also consider people who have limited internet access or lack experience using digital tools.

Recommendations

CareerGhana could begin by assessing whether students and job seekers would benefit from AI-supported career guidance. The company could then test a simple AI feature that helps users explore careers and identify the skills they need. It would also be important to protect users' personal information and ensure that AI-generated advice is accurate. Feedback from students and job seekers would help the company find out whether the new features are actually helping users before expanding it.

Conclusion

AI has the potential to improve education and career development in Ghana. Research suggests that it can support learning and help students develop useful skills, but responsible implementation is essential. CareerGhana could explore using AI to provide career guidance and help young people identify skills they need for employment. By focusing on accuracy, privacy, and accessibility, the company could use AI to better support students and job seekers.